UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 13: CLUSTERING

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

Prepared by:
Dr. Noor Jannah Zakaria
noor.jannah@utm.my

Innovating Solutions

Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Applications
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

Introduction

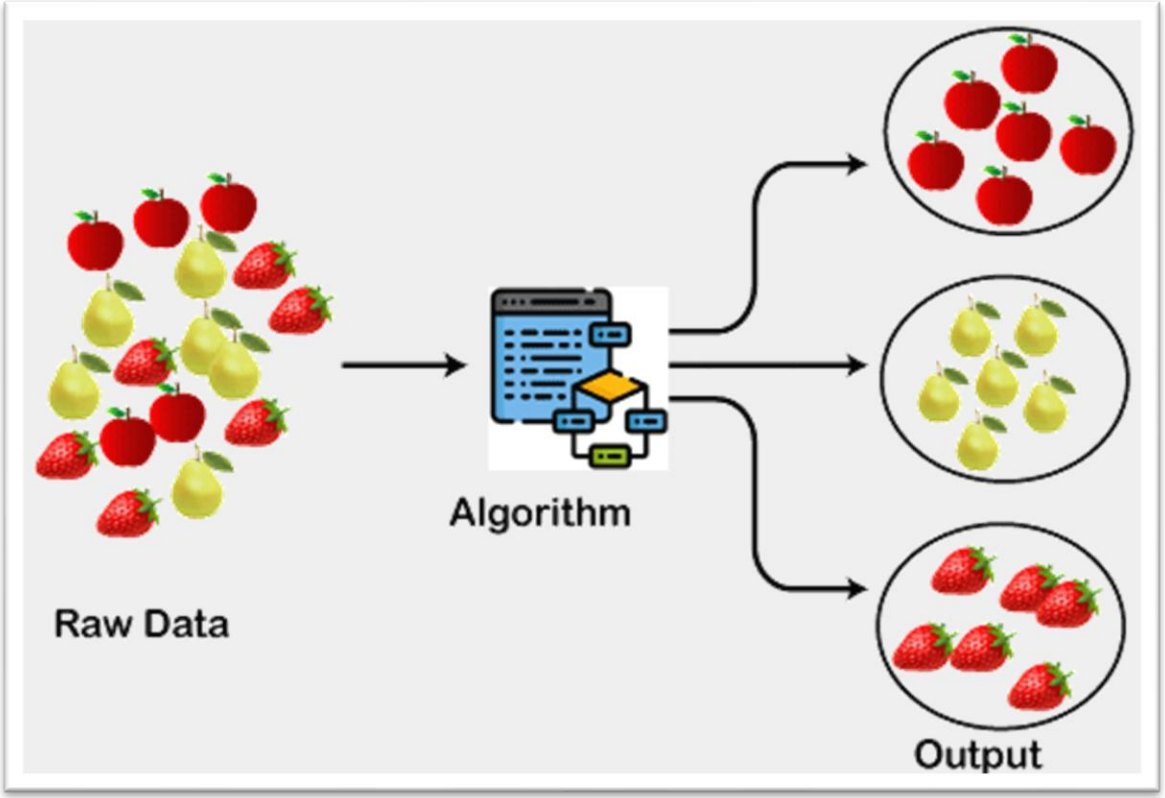
- Clustering or cluster analysis is a machine learning technique, which groups the unlabelled dataset. It can be defined as ***"A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group."***

- It does it by finding some similar patterns in the unlabelled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns.
- It is an unsupervised learning method; hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.
- After applying this clustering technique, each cluster or group is provided with a cluster-ID. ML system can use this id to simplify the processing of large and complex datasets.
- The clustering technique is commonly used for **statistical data analysis.**

www.utm.my

Introduction

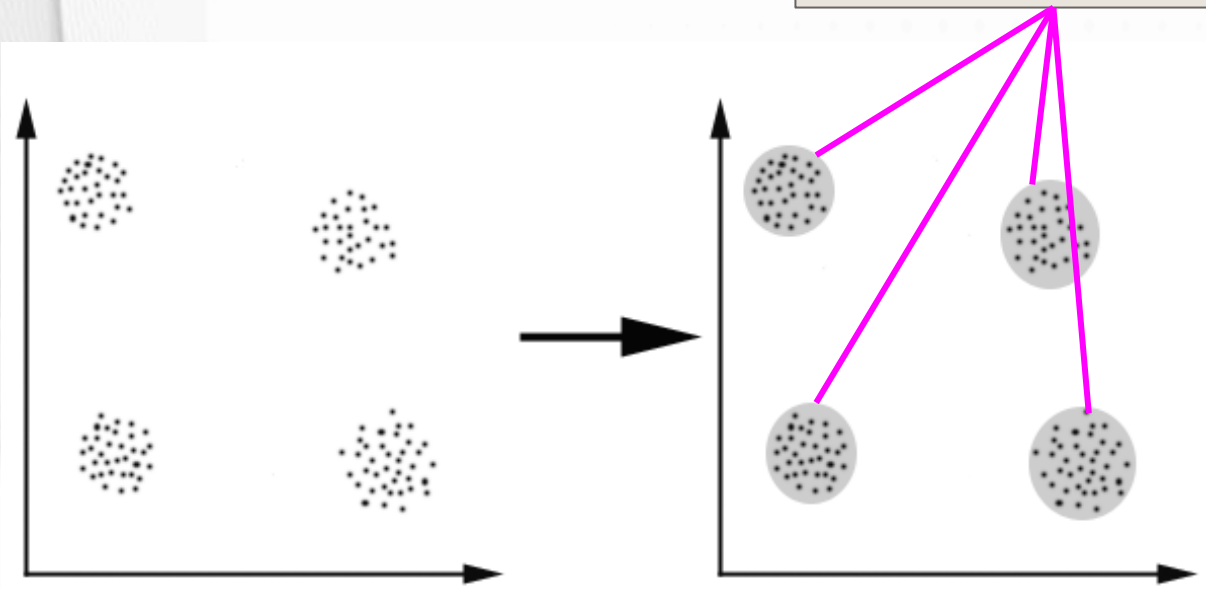
The below diagram explains the working of the clustering algorithm. We can see the different fruits are divided into several groups with similar properties.



www.utm.my

Introduction

There are four clusters identified in which the data can be divided.



The similarity criterion is distance: two or more objects belong to the same cluster if they are "close" according to a given distance.

This is known as distance-based clustering.

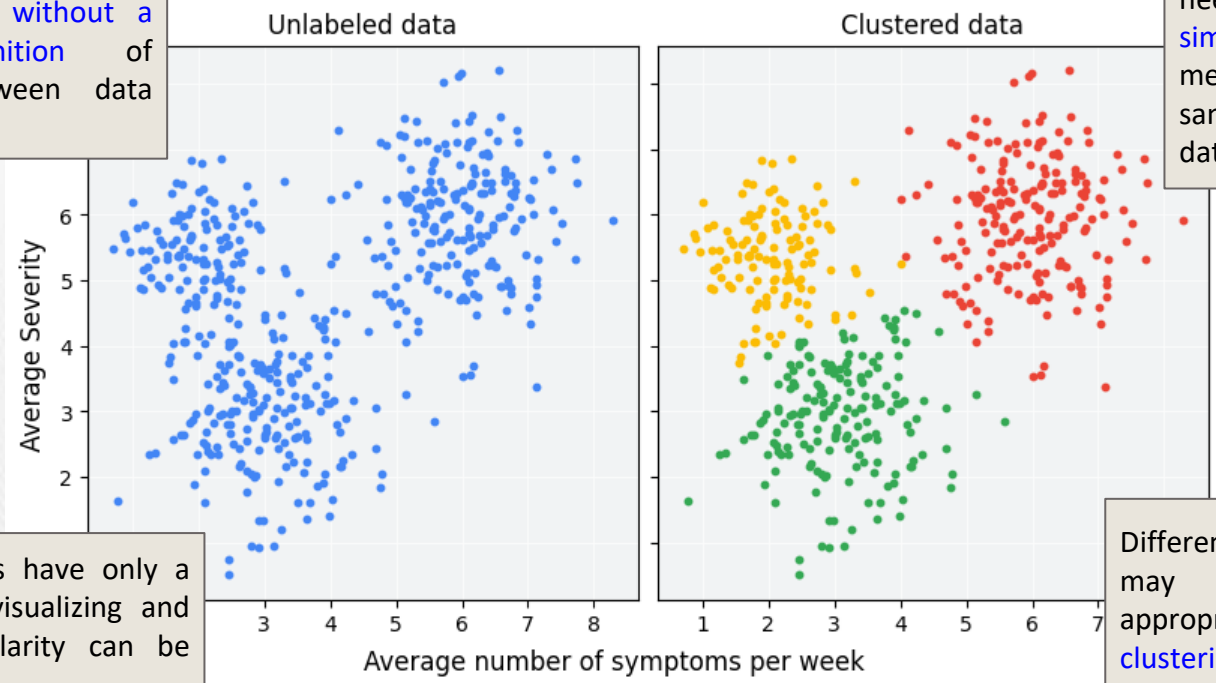
Conceptual clustering is where two or more objects are grouped based on their fit to descriptive concepts.

Introduction

- For example, a hypothetical patient study was designed to evaluate a new treatment protocol.
- In the study, patients report how many times per week they experience symptoms and the severity of the symptoms.
- Clustering analysis is used to group patients with similar treatment responses into clusters.

Introduction

The data forms three clusters, even without a formal definition of similarity between data points.



In real world applications, you need to explicitly define a similarity measure, or the metric used to compare samples, in terms of the dataset's features.

When examples have only a few features, visualizing and measuring similarity can be straightforward.

Different similarity measures may be more or less appropriate for different clustering scenarios.

Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Applications
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

Clustering Algorithms

- Machine learning datasets may have millions of examples and some clustering algorithms may not scale efficiently.
- Many clustering algorithms compute the similarity between all pair of examples. Their runtime increases as the square of the number of examples n, denoted as $O(n^2)$ in complexity notation.
- The $O(n^2)$ algorithms are not practical for datasets with millions of examples.
- The k-means algorithm has a complexity of $O(n)$, which means the algorithm scales linearly with n.

Application of Clustering

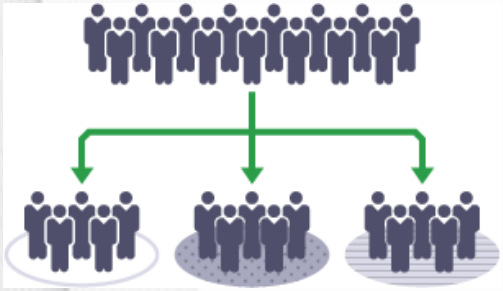
The clustering technique can be widely used in various tasks. Some most common uses of this technique are:

- Market Segmentation
- Statistical data analysis
- Social network analysis
- Image segmentation
- Anomaly detection, etc.

- Apart from these general usages, it is used by the **Amazon** in its recommendation system to provide the recommendations as per the past search of products. **Netflix** also uses this technique to recommend the movies and web-series to its users as per the watch history.

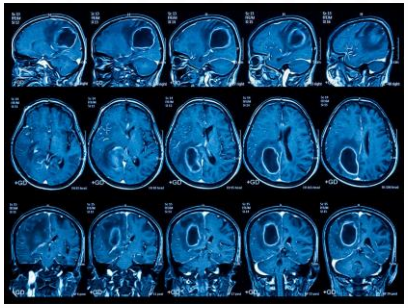
www.utm.my

Application of Clustering



Market segmentation

Image source: bbc.co.uk



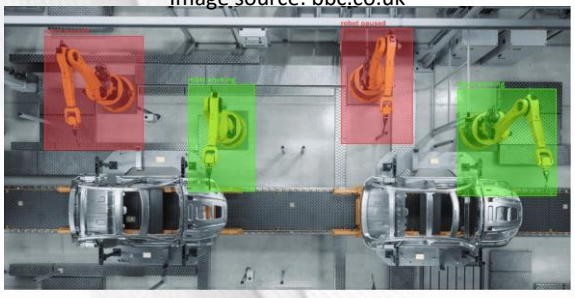
Medical imaging

Image source: knowledgetransfer



Social network analysis

Image source: medium.com



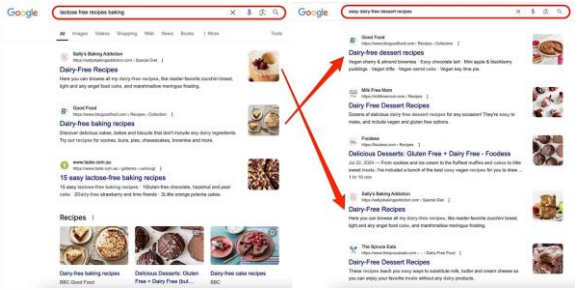
Anomaly detection

Image source: alwaysai.co



Image segmentation

Image source: indiaAI



Search result grouping

Image source: seranking.com

Application of Clustering

1. **In Identification of Cancer Cells:** The clustering algorithms are widely used for the identification of cancerous cells. It divides the cancerous and non-cancerous data sets into different groups.

2. **In Search Engines:** Search engines also work on the clustering technique. The search result appears based on the closest object to the search query. It does it by grouping similar data objects in one group that is far from the other dissimilar objects. The accurate result of a query depends on the quality of the clustering algorithm used.

3. **Customer Segmentation:** It is used in market research to segment the customers based on their choice and preferences.

4. **In Biology:** It is used in the biology stream to classify different species of plants and animals using the image recognition technique.

5. **In Land Use:** The clustering technique is used in identifying the area of similar lands use in the GIS database. This can be very useful to find that for what purpose the particular land should be used, that means for which purpose it is more suitable.

Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Applications
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

Types of Clustering Methods

The clustering methods are broadly divided into **Hard Clustering** (data point belongs to only one group) and **Soft Clustering** (data points can belong to another group also). But there are also other various approaches of Clustering exist. Below are the main clustering methods used in Machine learning:

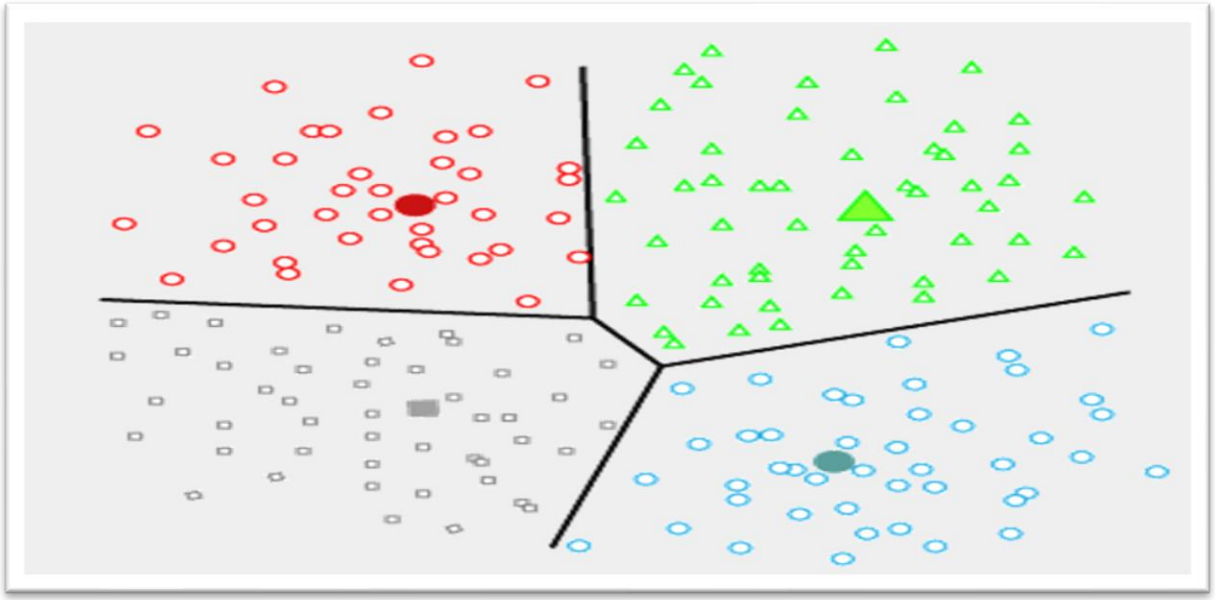
- Partitioning Clustering
- Density-Based Clustering
- Distribution Model-Based Clustering
- Hierarchical Clustering
- Fuzzy Clustering

www.utm.my

Types of Clustering Methods

1.Partitioning Clustering :

- It is a type of clustering that divides the data into non-hierarchical groups. It is also known as the **Centroid-based method**. The most common example of partitioning clustering is the **K-Means Clustering algorithm**.
- In this type, the dataset is divided into a set of k groups, where K is used to define the number of pre-defined groups. The cluster center is created in such a way that the distance between the data points of one cluster is minimum as compared to another cluster centroid.

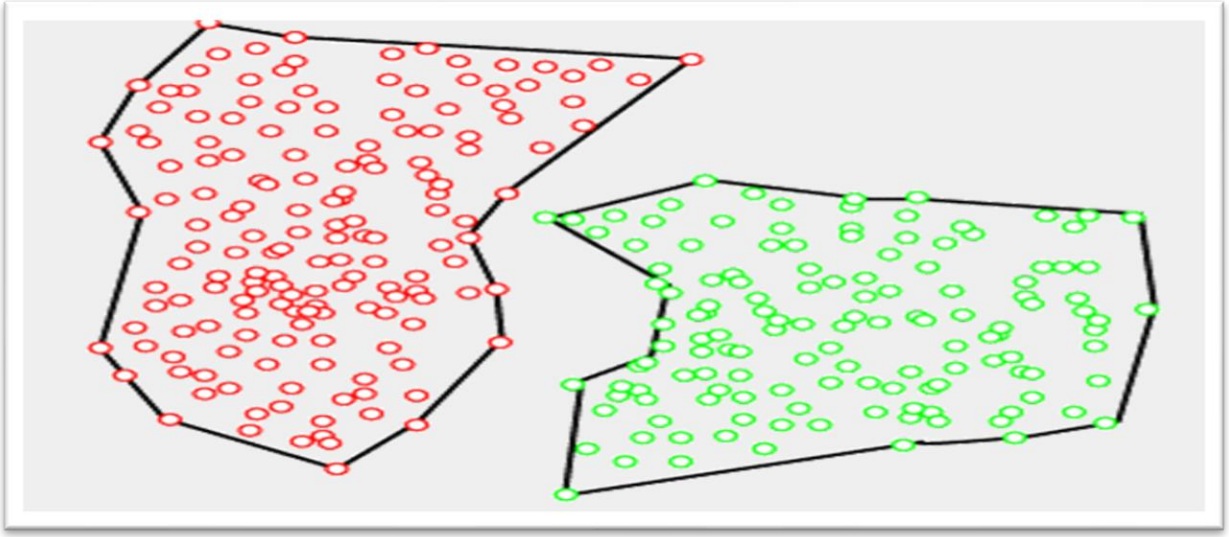


www.utm.my

Types of Clustering Methods

2.Density-Based Clustering :

- The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected. This algorithm does it by identifying different clusters in the dataset and connects the areas of high densities into clusters. The dense areas in data space are divided from each other by sparser areas.
- These algorithms can face difficulty in clustering the data points if the dataset has varying densities and high dimensions.

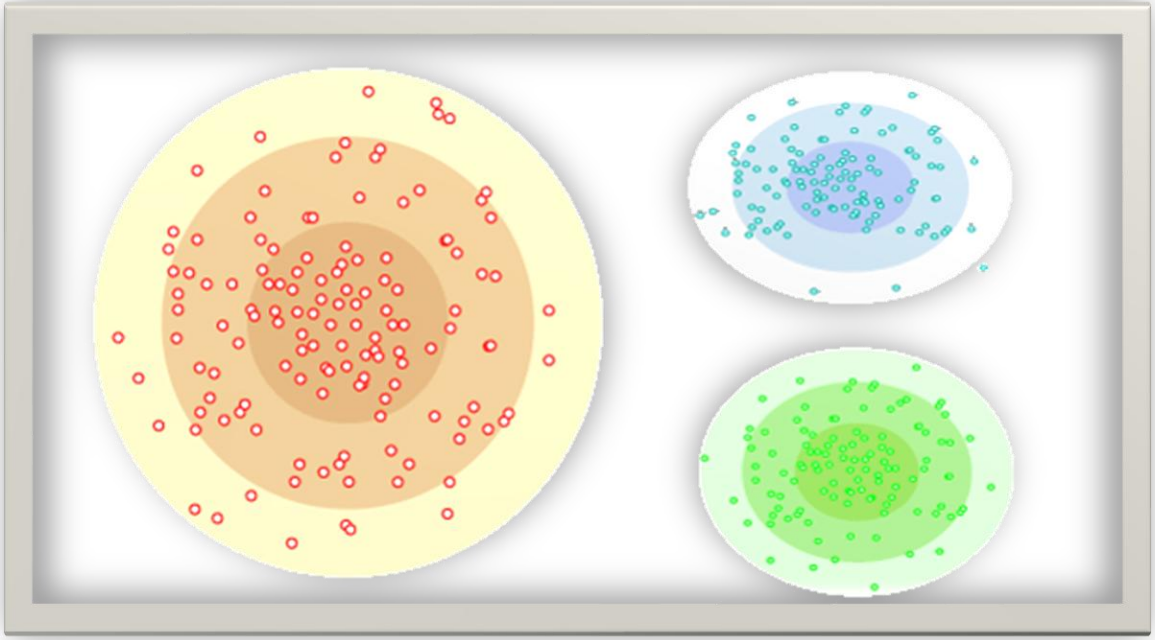


www.utm.my

Types of Clustering Methods

3.Distribution Model-Based Clustering :

- In the distribution model-based clustering method, the data is divided based on the probability of how a dataset belongs to a particular distribution. The grouping is done by assuming some distributions commonly **Gaussian Distribution**.
- The example of this type is the **Expectation-Maximization Clustering algorithm** that uses Gaussian Mixture Models (GMM).

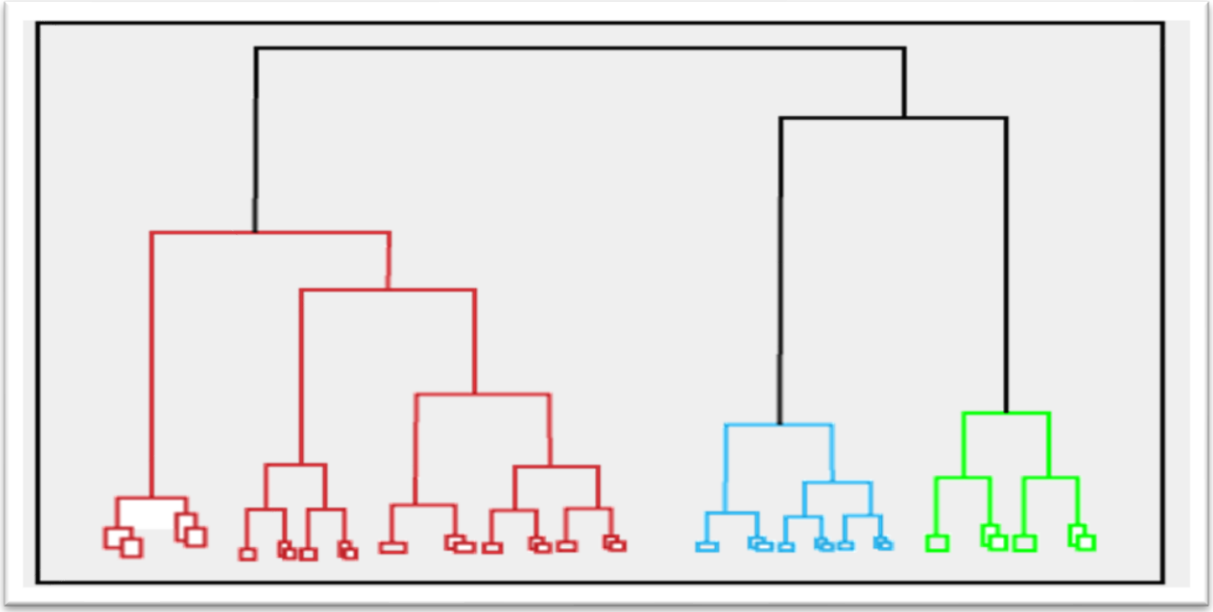


www.utm.my

Types of Clustering Methods

4.Hierarchical Clustering :

- Hierarchical clustering can be used as an alternative for the partitioned clustering as there is no requirement of pre-specifying the number of clusters to be created. In this technique, the dataset is divided into clusters to create a tree-like structure, which is also called a **dendrogram**. The observations or any number of clusters can be selected by cutting the tree at the correct level. The most common example of this method is the **Agglomerative Hierarchical algorithm**.



www.utm.my

Hierarchical Clustering

- Hierarchical clustering is another unsupervised machine learning algorithm, which is used to group the unlabeled datasets into a cluster and also known as **hierarchical cluster analysis** or **HCA.**
- In this algorithm, we develop the hierarchy of clusters in the form of a tree, and this tree-shaped structure is known as the **dendrogram**.
- Sometimes the results of K-means clustering and hierarchical clustering may look similar, but they both differ depending on how they work. As there is no requirement to predetermine the number of clusters as we did in the K-Means algorithm.

The hierarchical clustering technique has two approaches:

- **Agglomerative:** Agglomerative is a **bottom-up** approach, in which the algorithm starts with taking all data points as single clusters and merging them until one cluster is left.
- **Divisive:** Divisive algorithm is the reverse of the agglomerative algorithm as it is a **top-down approach.**

www.utm.my

Hierarchical Clustering

1.Agglomerative Hierarchical clustering :

- The agglomerative hierarchical clustering algorithm is a popular example of HCA.

- To group the datasets into clusters, it follows the **bottom-up approach**. It means, this algorithm considers each dataset as a single cluster at the beginning, and then start combining the closest pair of clusters together. It does this until all the clusters are merged into a single cluster that contains all the datasets.

- This hierarchy of clusters is represented in the form of the dendrogram.

www.utm.my

Hierarchical Clustering

Working of Agglomerative Hierarchical clustering :

The working of the AHC algorithm can be explained using the below steps:

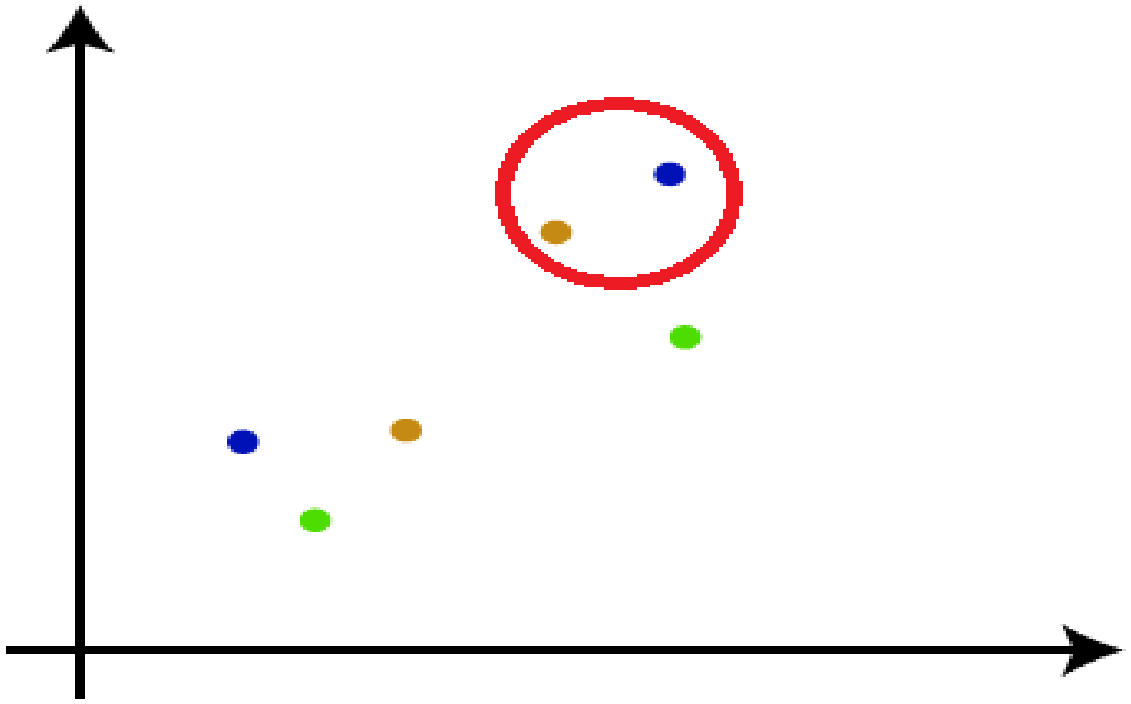
Step-1: Create each data point as a single cluster. Let's say there are N data points, so the number of clusters will also be N.



www.utm.my

Hierarchical Clustering

Step-2: Take two closest data points or clusters and merge them to form one cluster. So, there will now be N-1 clusters.



www.utm.my

Hierarchical Clustering

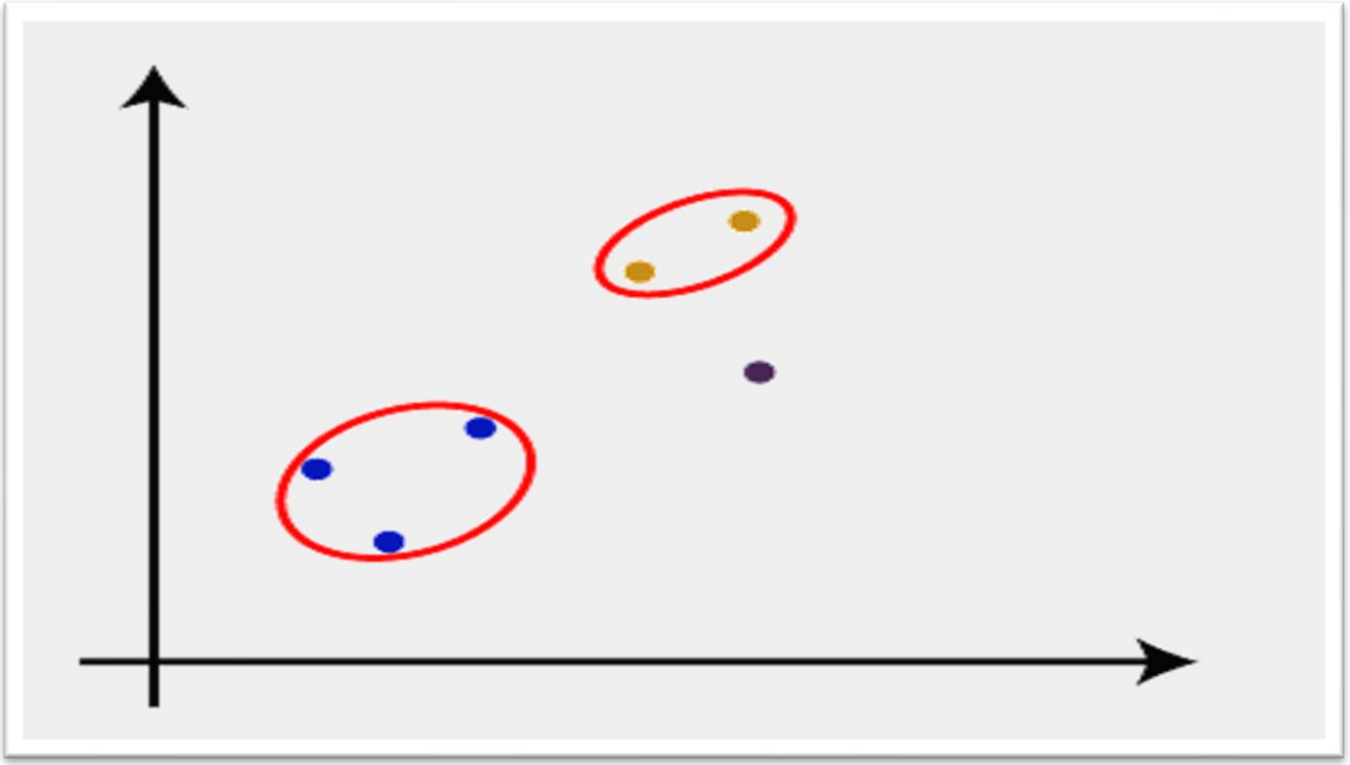
Step-3: Again, take the two closest clusters and merge them together to form one cluster. There will be N-2 clusters.



www.utm.my

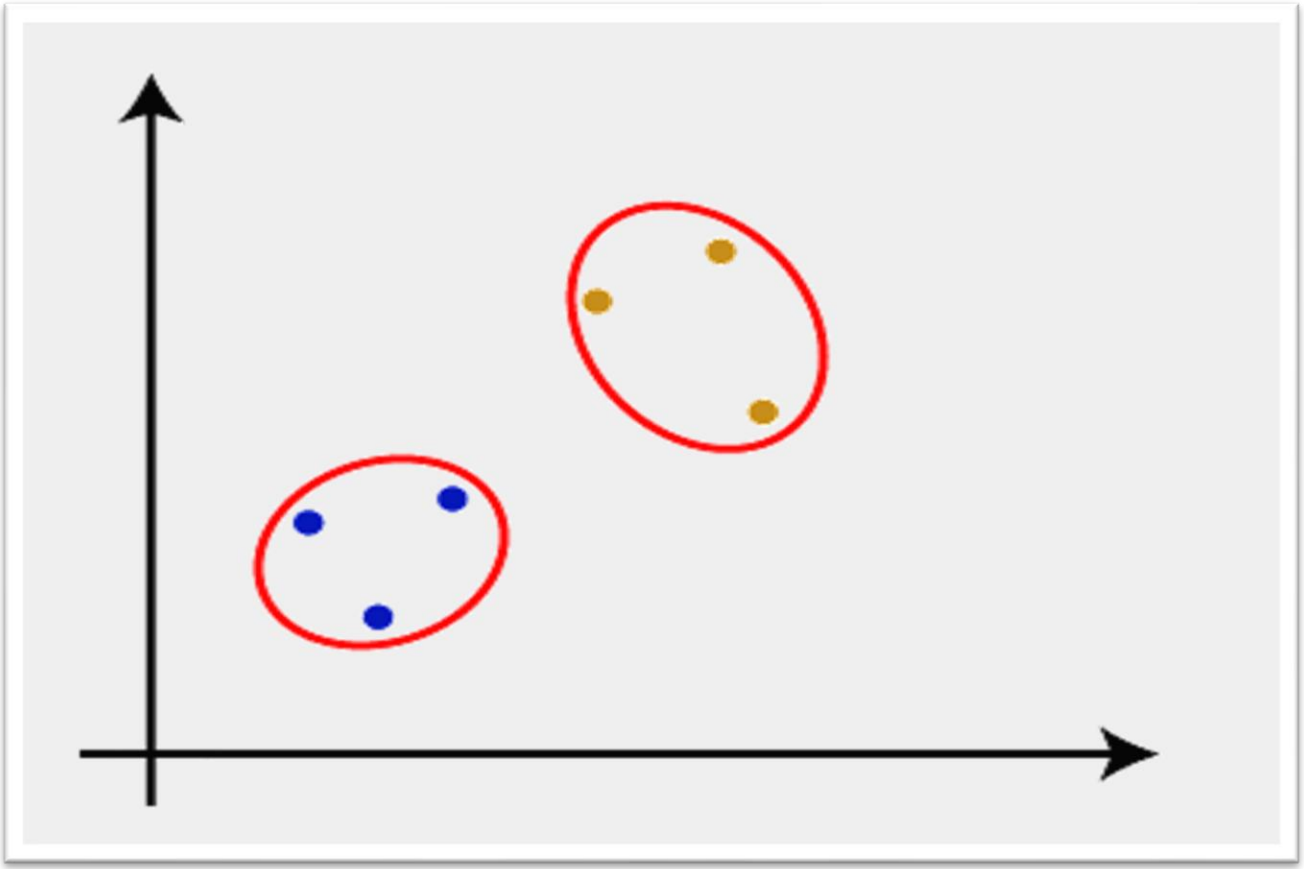
Hierarchical Clustering

Step-4: Repeat Step 3 until only one cluster left. So, we will get the following clusters. Consider the below images:



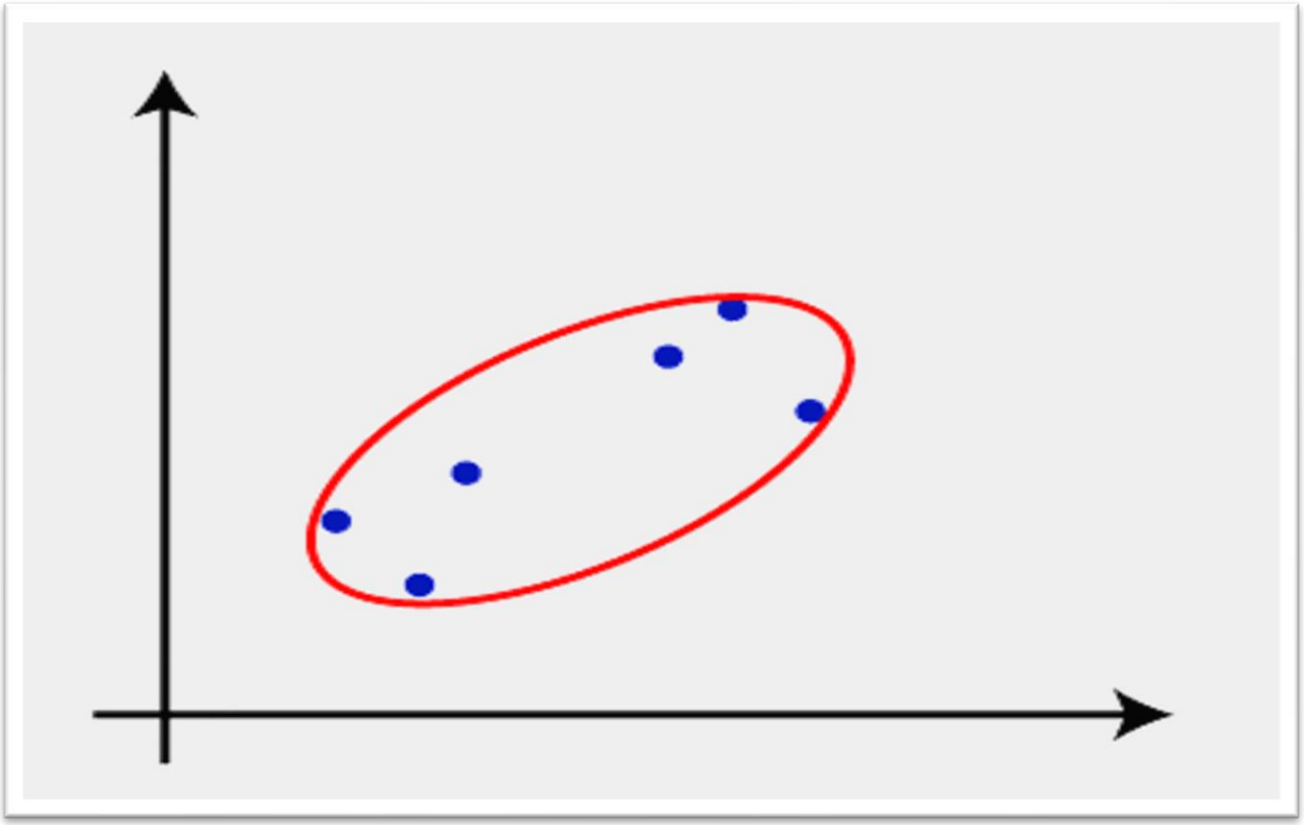
www.utm.my

Hierarchical Clustering



www.utm.my

Hierarchical Clustering



Step-5: Once all the clusters are combined into one big cluster, develop the dendrogram to divide the clusters as per the problem.

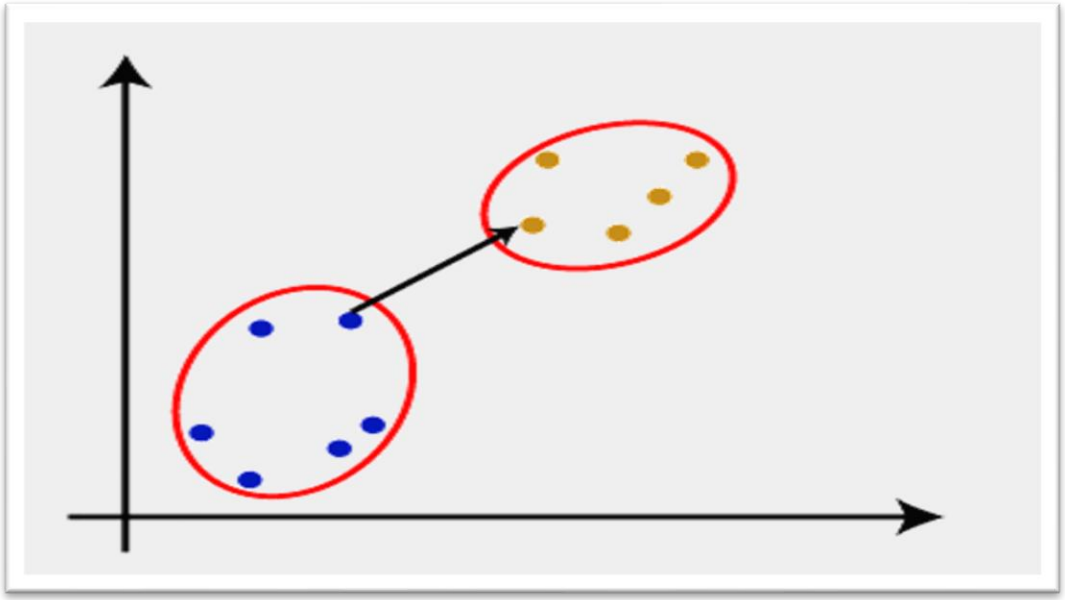
www.utm.my

Hierarchical Clustering

Measure for the distance between two clusters :

- As we have seen, the **closest distance** between the two clusters is crucial for the hierarchical clustering. There are various ways to calculate the distance between two clusters, and these ways decide the rule for clustering. These measures are called **Linkage methods**. Some of the popular linkage methods are given below:

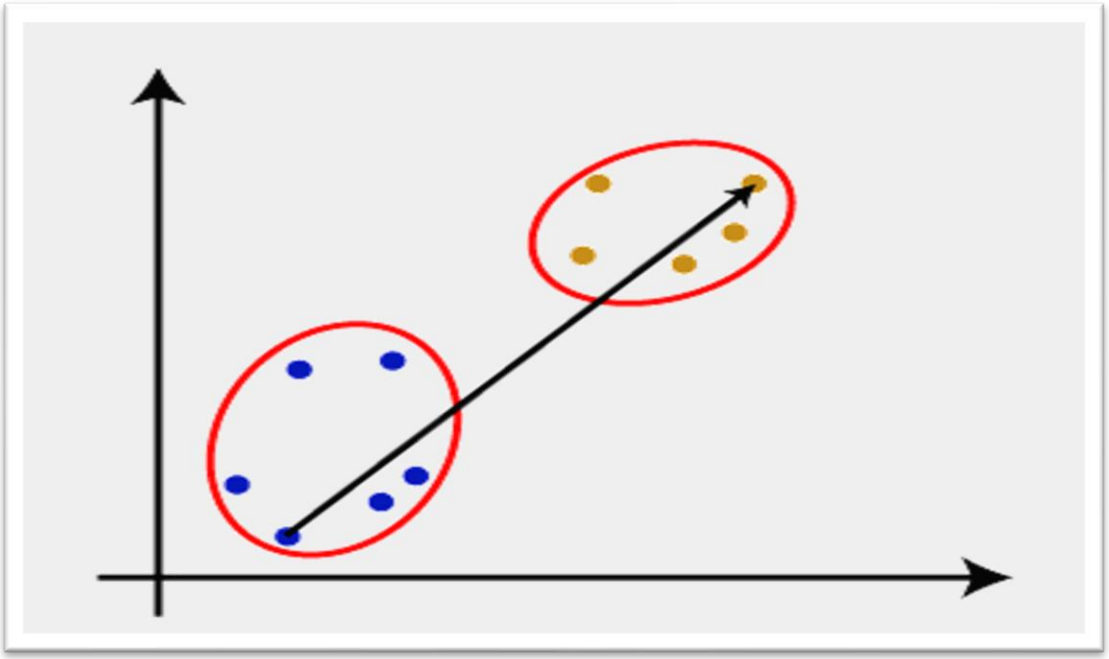
  1.Single Linkage: It is the Shortest Distance between the closest points of the clusters. Consider the below image:



www.utm.my

Hierarchical Clustering

2. **Complete Linkage:** It is the farthest distance between the two points of two different clusters. It is one of the popular linkage methods as it forms tighter clusters than single-linkage.

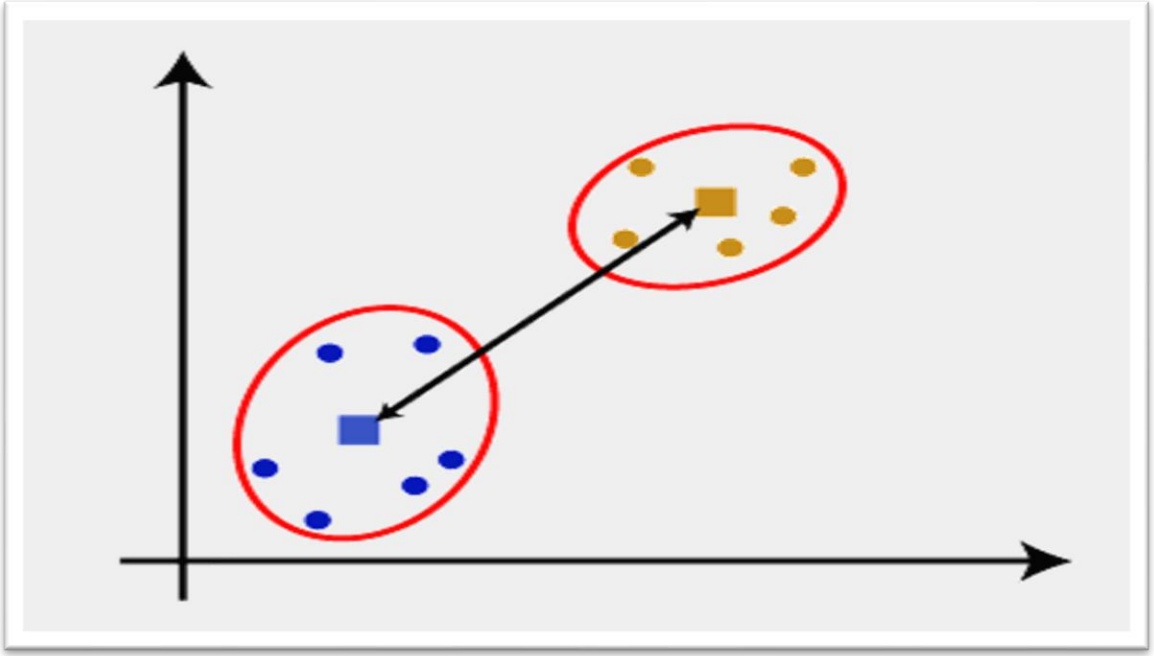


www.utm.my

Hierarchical Clustering

3.Average Linkage: It is the linkage method in which the distance between each pair of datasets is added up and then divided by the total number of datasets to calculate the average distance between two clusters. It is also one of the most popular linkage methods.

4.Centroid Linkage: It is the linkage method in which the distance between the centroid of the clusters is calculated. Consider the below image:

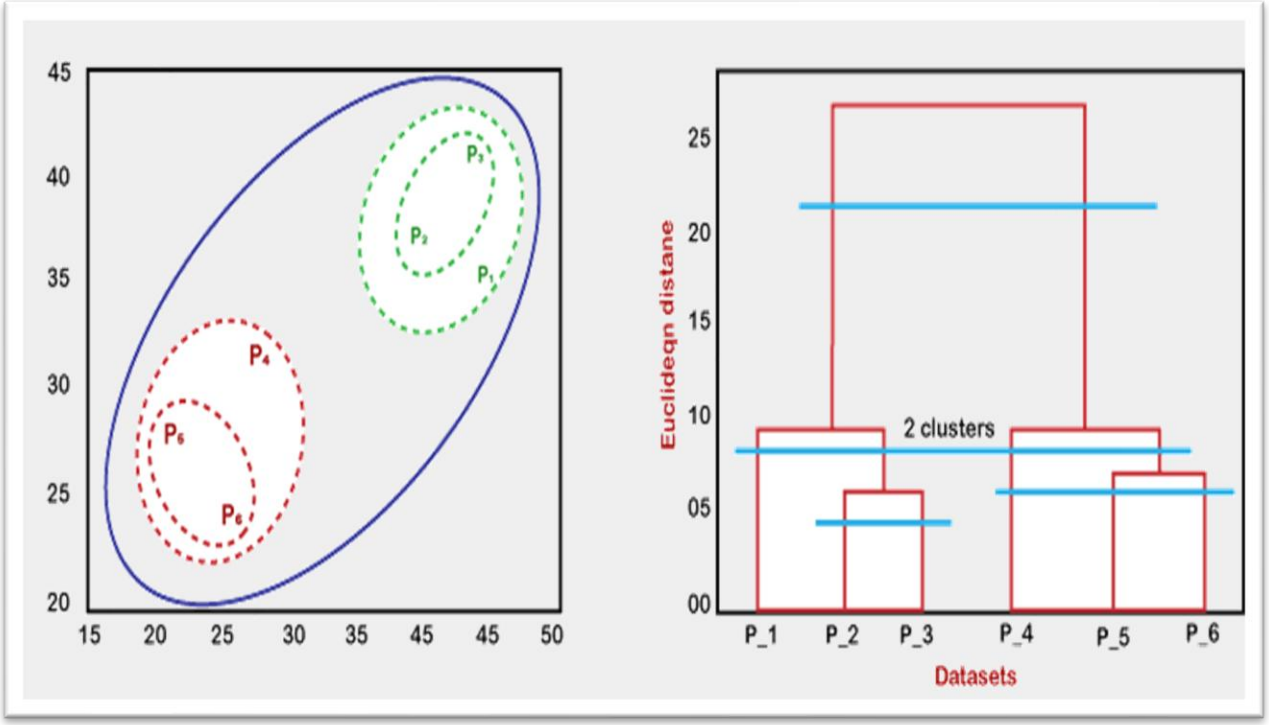


www.utm.my

Hierarchical Clustering

Working of Dendrogram in Hierarchical clustering :

- The dendrogram is a tree-like structure that is mainly used to store each step as a memory that the HC algorithm performs. In the dendrogram plot, the Y-axis shows the Euclidean distances between the data points, and the x-axis shows all the data points of the given dataset.



www.utm.my

Hierarchical Clustering

- In the above diagram, the left part is showing how clusters are created in agglomerative clustering, and the right part is showing the corresponding dendrogram.
- Firstly, the data points P2 and P3 combine together and form a cluster, correspondingly a dendrogram is created, which connects P2 and P3 with a rectangular shape. The height is decided according to the Euclidean distance between the data points.
- In the next step, P5 and P6 form a cluster, and the corresponding dendrogram is created. It is higher than of previous, as the Euclidean distance between P5 and P6 is a little bit greater than the P2 and P3.
- Again, two new dendrogram are created that combine P1, P2, and P3 in one dendrogram, and P4, P5, and P6, in another dendrogram.
- At last, the final dendrogram is created that combines all the data points together.
- We can cut the dendrogram tree structure at any level as per our requirement.

www.utm.my

Types of Clustering Methods

5.Fuzzy Clustering :

- Fuzzy clustering is a type of soft method in which a data object may belong to more than one group or cluster. Each dataset has a set of membership coefficients, which depend on the degree of membership to be in a cluster. **Fuzzy C-means algorithm** is the example of this type of clustering; it is sometimes also known as the Fuzzy k-means algorithm.

www.utm.my

Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Application
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

K-Means Clustering

- K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, as if K=2, there will be two clusters, and for K=3, there will be three clusters, and so on.

- It allows us to cluster the data into different groups and a convenient way to discover the categories of groups in the unlabeled dataset on its own without the need for any training.

- It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

- The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm.

www.utm.my

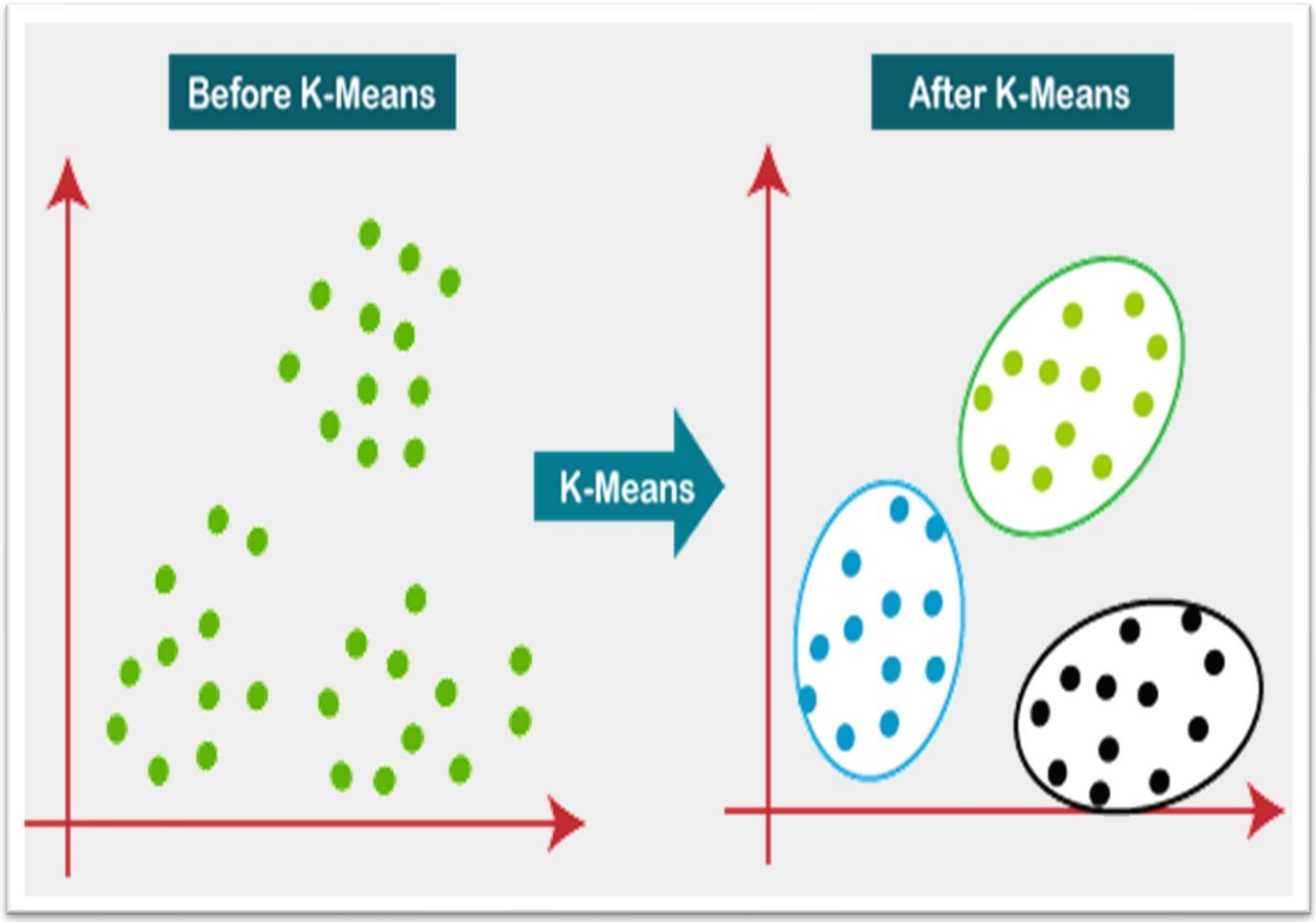
K-Means Clustering

The k-means clustering algorithm mainly performs two tasks:

- Determines the best value for K center points or centroids by an iterative process.
- Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.
- Hence each cluster has data points with some commonalities, and it is away from other clusters. The below diagram explains the working of the K-means Clustering Algorithm:

www.utm.my

K-Means Clustering



www.utm.my

K-Means Clustering

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

www.utm.my

K-Means Clustering

Suppose we have two variables M1 and M2. The x-y axis scatter plot of these two variables is given below:



www.utm.my

K-Means Clustering

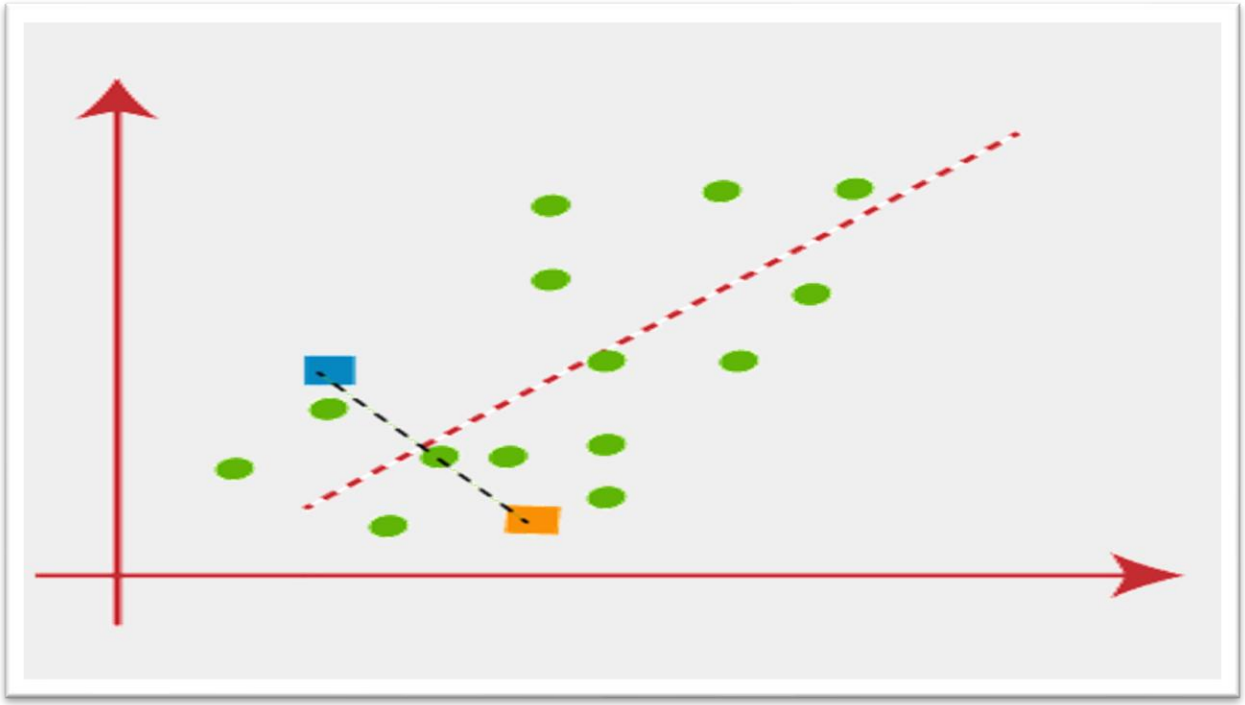
- Let's take number k of clusters, i.e., K=2, to identify the dataset and to put them into different clusters. It means here we will try to group these datasets into two different clusters.
- We need to choose some random k points or centroid to form the cluster. These points can be either the points from the dataset or any other point. So, here we are selecting the below two points as k points, which are not the part of our dataset. Consider the below image:



www.utm.my

K-Means Clustering

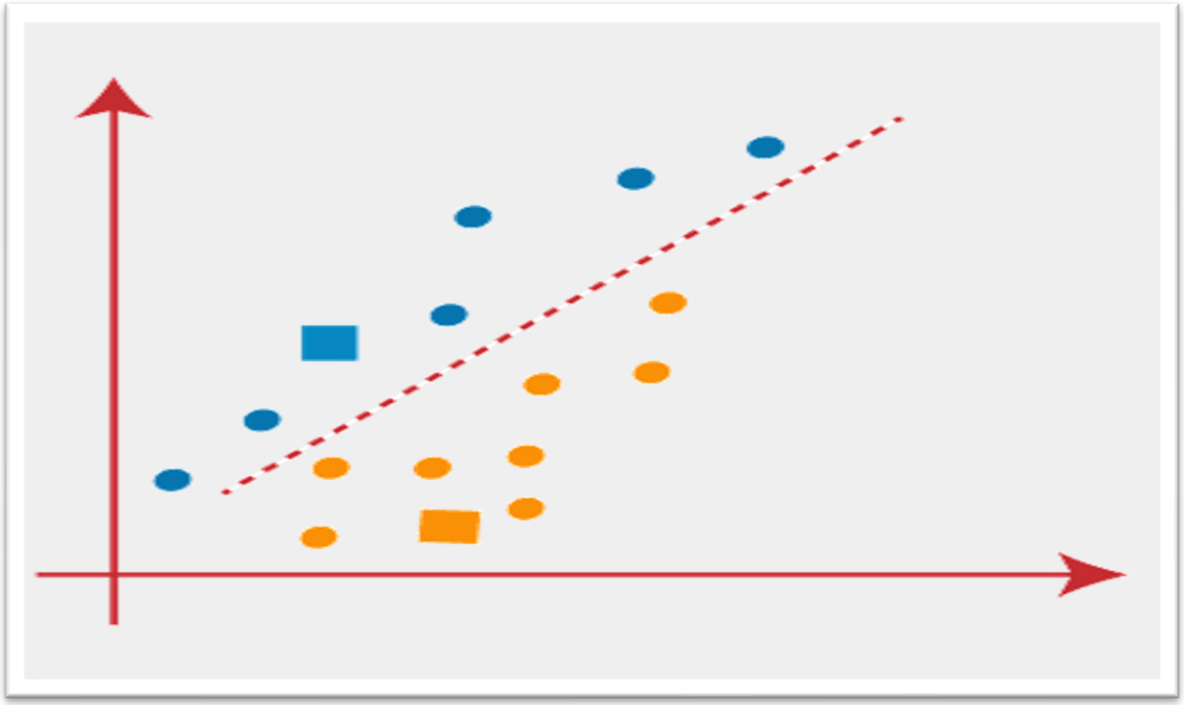
Now we will assign each data point of the scatter plot to its closest K-point or centroid. We will compute it by applying some mathematics that we have studied to calculate the distance between two points. So, we will draw a median between both the centroids. Consider the below image:



www.utm.my

K-Means Clustering

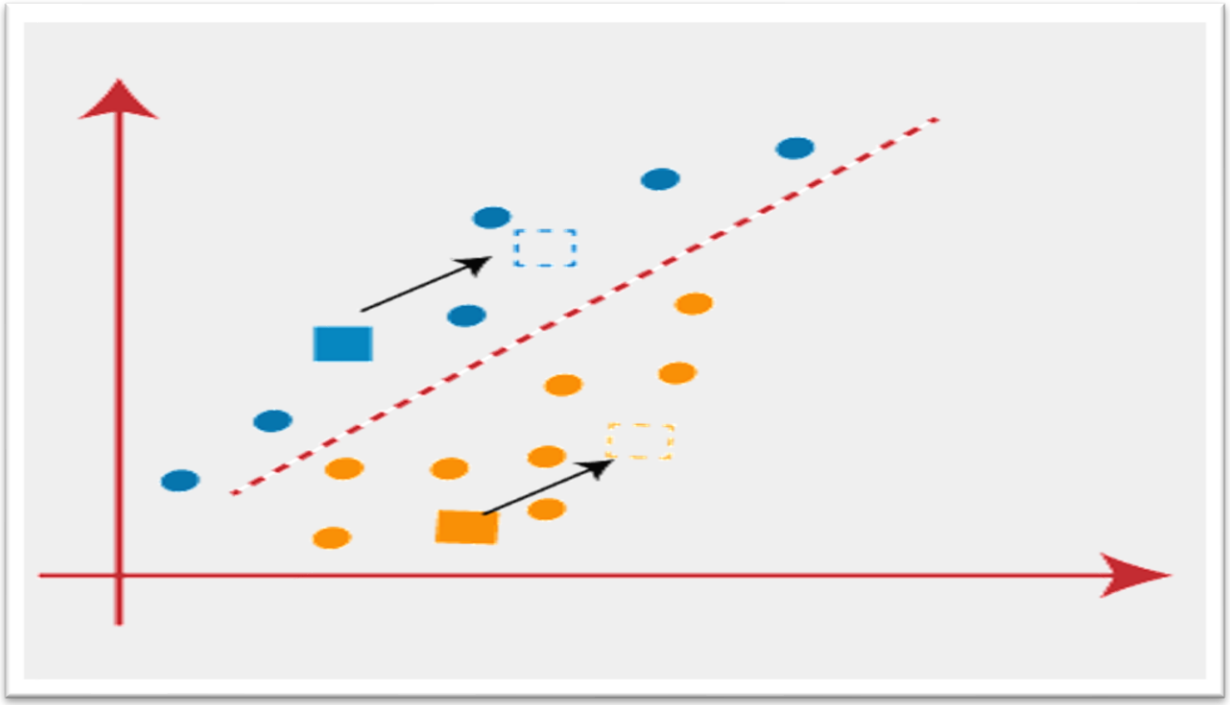
- From the above image, it is clear that points left side of the line is near to the K1 or blue centroid, and points to the right of the line are close to the yellow centroid. Let's color them as blue and yellow for clear visualization.



www.utm.my

K-Means Clustering

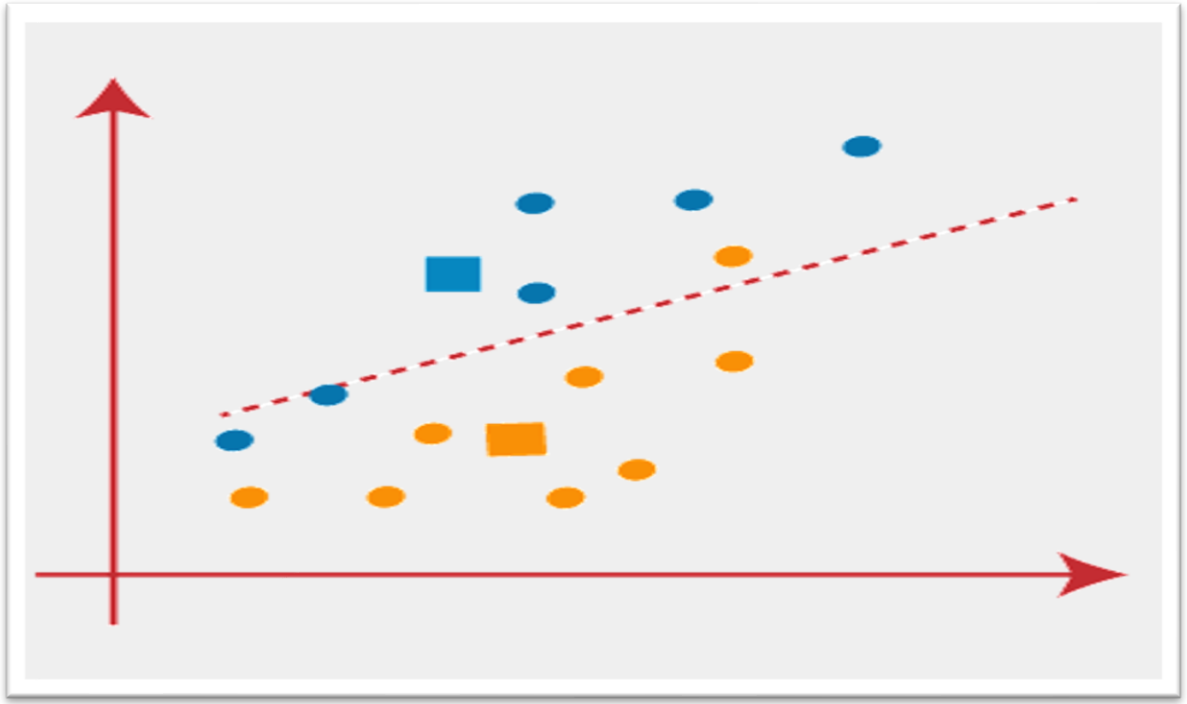
- As we need to find the closest cluster, so we will repeat the process by choosing **a new centroid**. To choose the new centroids, we will compute the center of gravity of these centroids, and will find new centroids as below:



www.utm.my

K-Means Clustering

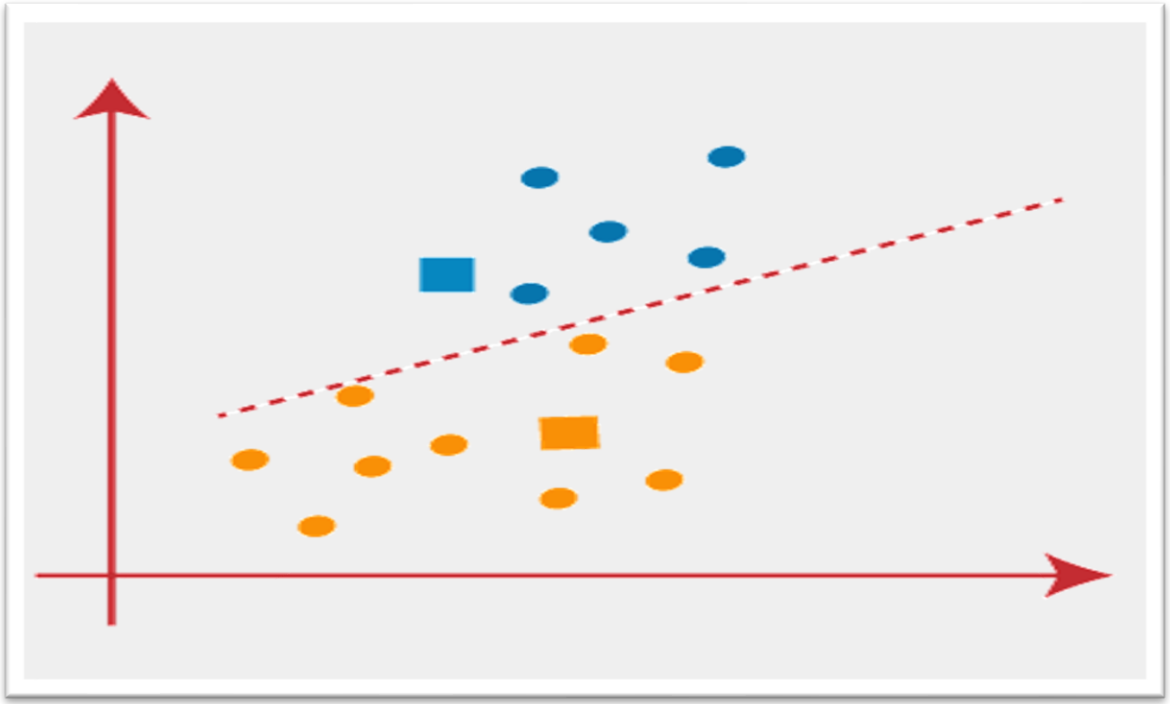
- Next, we will reassign each datapoint to the new centroid. For this, we will repeat the same process of finding a median line. The median will be like below image:



www.utm.my

K-Means Clustering

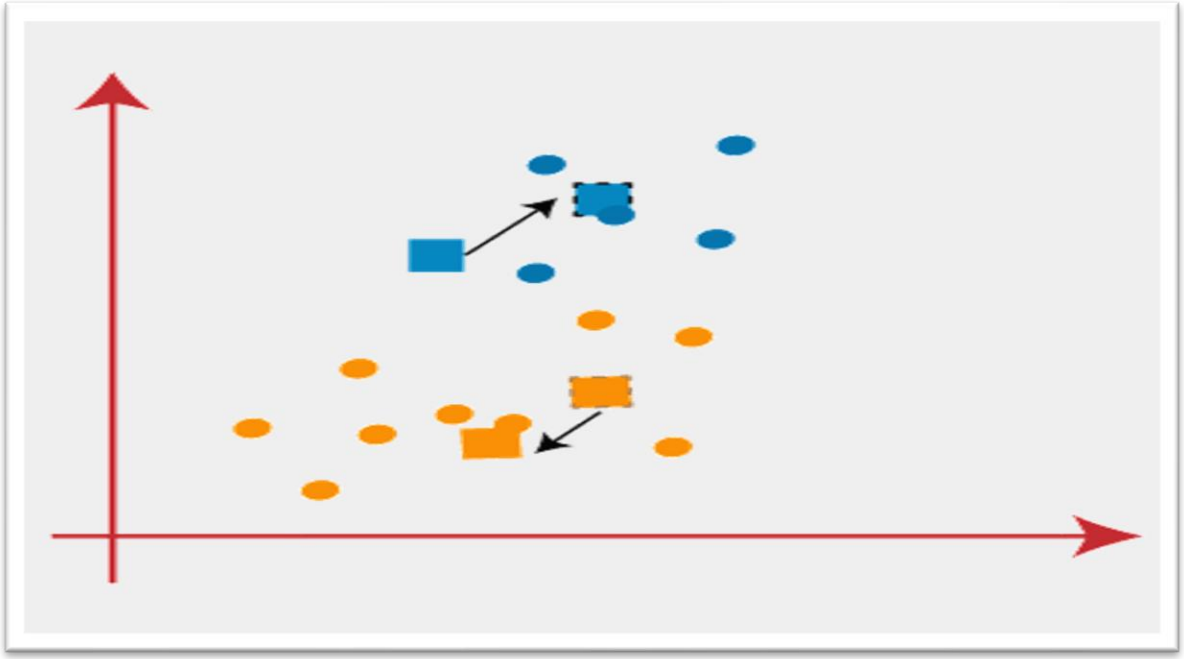
- From the above image, we can see, one yellow point is on the left side of the line, and two blue points are right to the line. So, these three points will be assigned to new centroids.



www.utm.my

K-Means Clustering

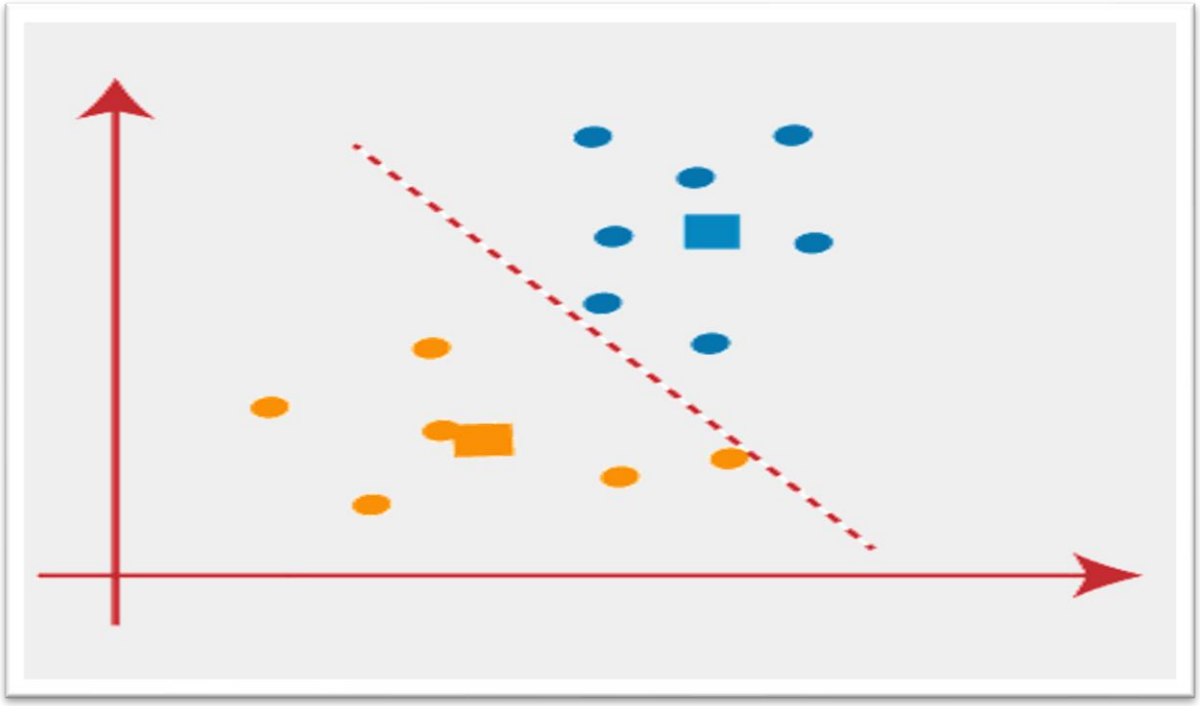
- As reassignment has taken place, so we will again go to the step-4, which is finding new centroids or K-points.
- We will repeat the process by finding the center of gravity of centroids, so the new centroids will be as shown in the below image:



www.utm.my

K-Means Clustering

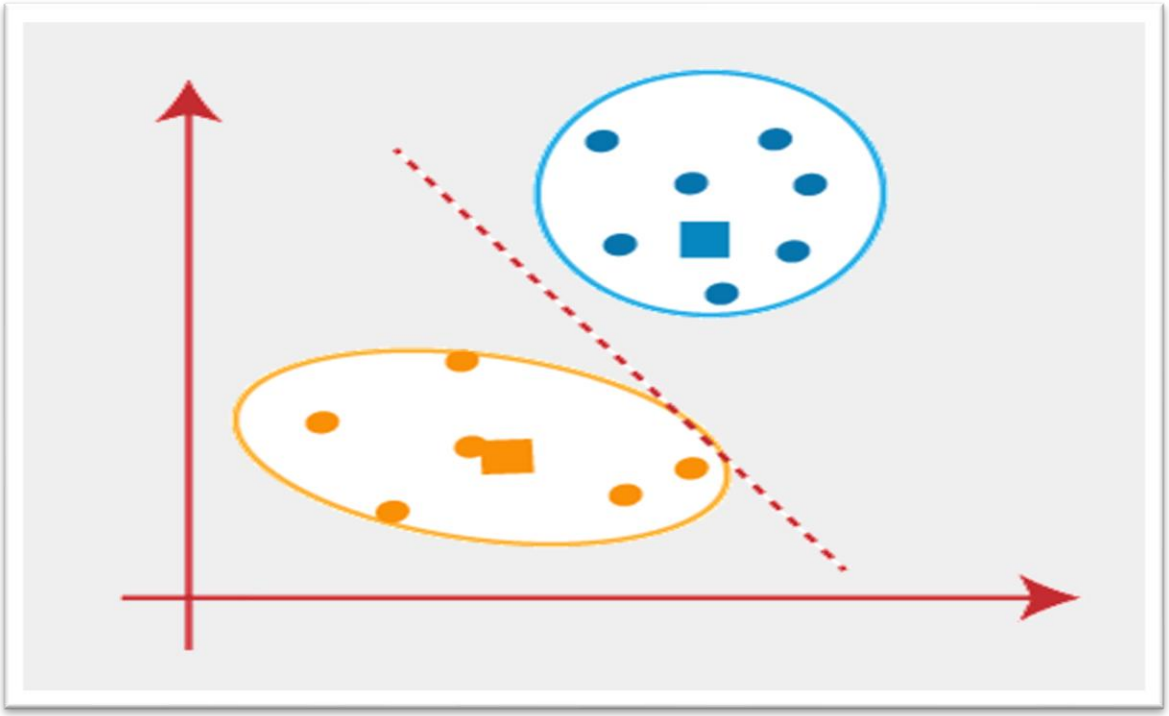
- As we got the new centroids so again will draw the median line and reassign the data points. So, the image will be:



www.utm.my

K-Means Clustering

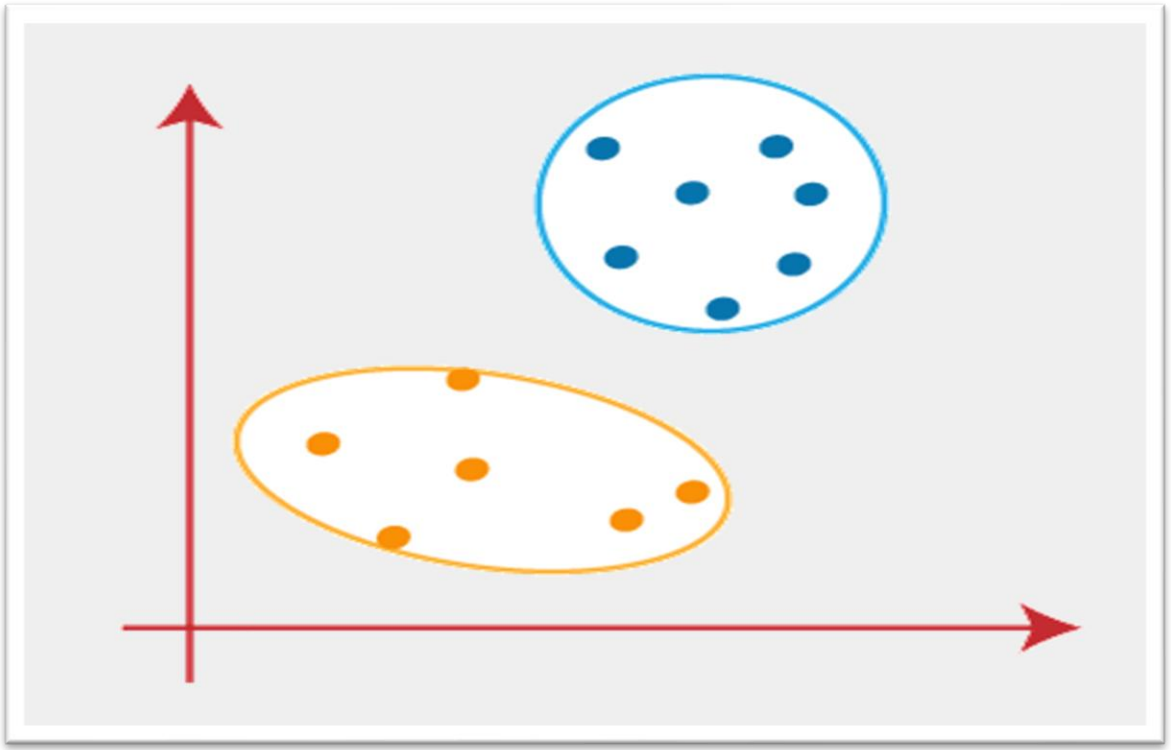
- We can see in the above image; there are no dissimilar data points on either side of the line, which means our model is formed. Consider the below image:



www.utm.my

K-Means Clustering

- As our model is ready, so we can now remove the assumed centroids, and the two final clusters will be as shown in the below image:



www.utm.my

K-Means Clustering

Choose the value of "K number of clusters" in K-means Clustering :

The performance of the K-means clustering algorithm depends upon highly efficient clusters that it forms. But choosing the optimal number of clusters is a big task. There are some different ways to find the optimal number of clusters, but here we are discussing the most appropriate method to find the number of clusters or value of K. The method is given below:

Elbow Method :

The Elbow method is one of the most popular ways to find the optimal number of clusters. This method uses the concept of WCSS value. **WCSS** stands for **Within Cluster Sum of Squares**, which defines the total variations within a cluster. The formula to calculate the value of WCSS (for 3 clusters) is given below:

www.utm.my

K-Means Clustering

$$WCSS = \sum_{Pi \text{ in Cluster1}} distance(P_i\ C_1)^2 + \sum_{Pi \text{ in Cluster2}} distance(P_i\ C_2)^2 + \sum_{Pi \text{ in CLuster3}} distance(P_i\ C_3)^2$$

In the above formula of WCSS,

- $\sum_{Pi \text{ in Cluster1}} distance(P_i\ C_1)^2$: It is the sum of the square of the distances between each data point and its centroid within a cluster1 and the same for the other two terms.
- To measure the distance between data points and centroid, we can use any method such as Euclidean distance or Manhattan distance.
- To find the optimal value of clusters, the elbow method follows the below steps:
- It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).

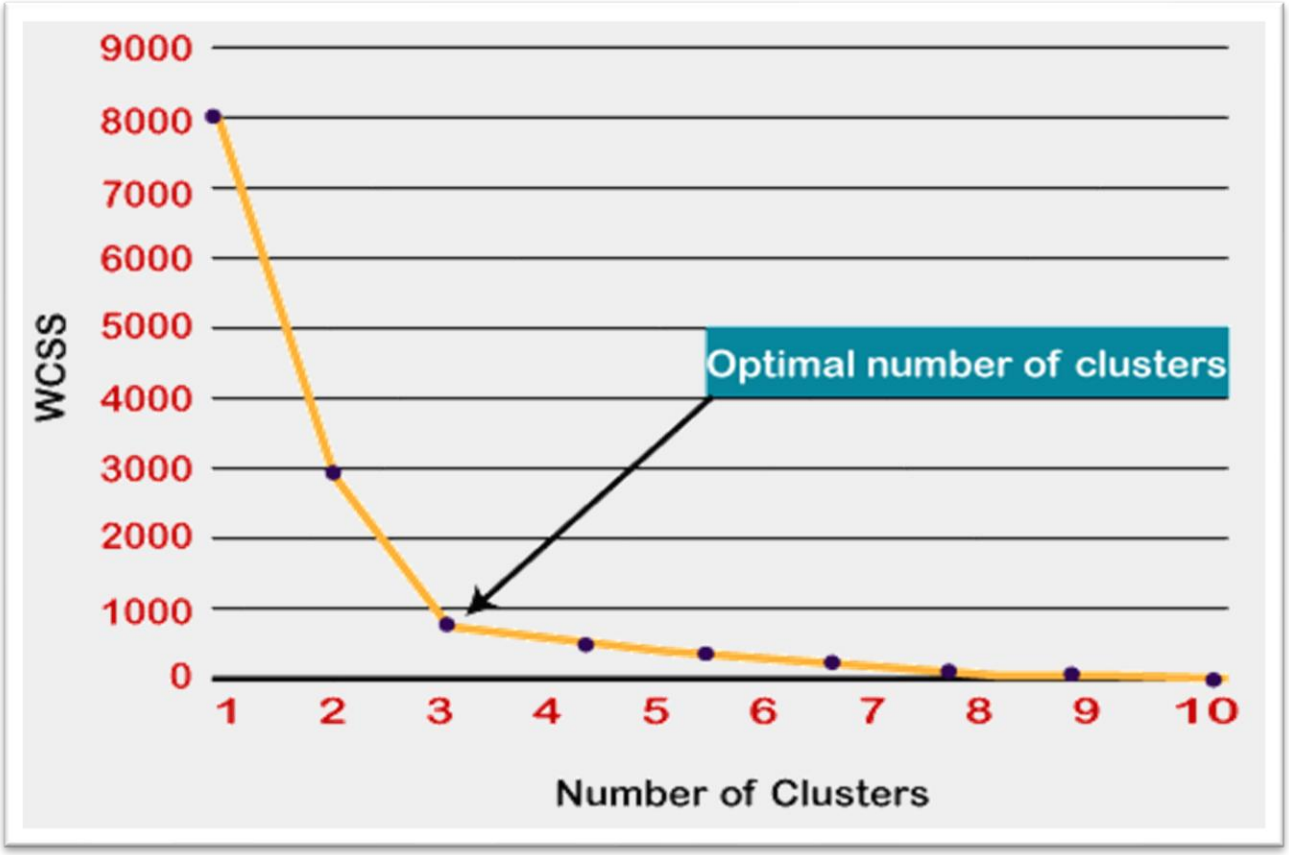
www.utm.my

K-Means Clustering

- For each value of K, calculates the WCSS value.
- Plots a curve between calculated WCSS values and the number of clusters K.
- The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.
- Since the graph shows the sharp bend, which looks like an elbow, hence it is known as the elbow method. The graph for the elbow method looks like the below image:

www.utm.my

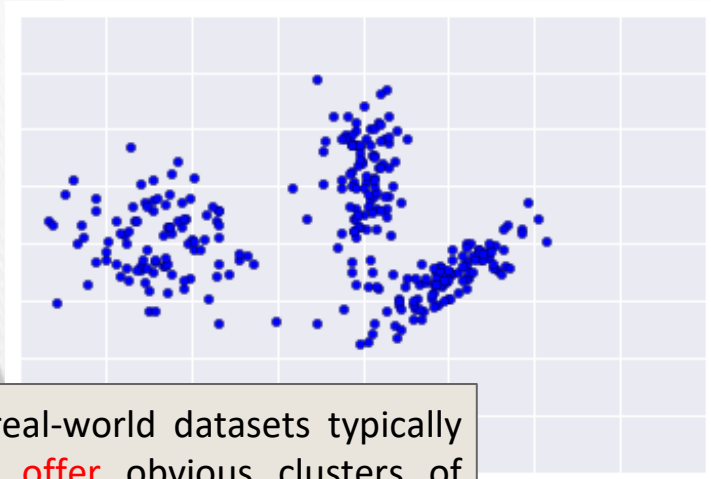
K-Means Clustering



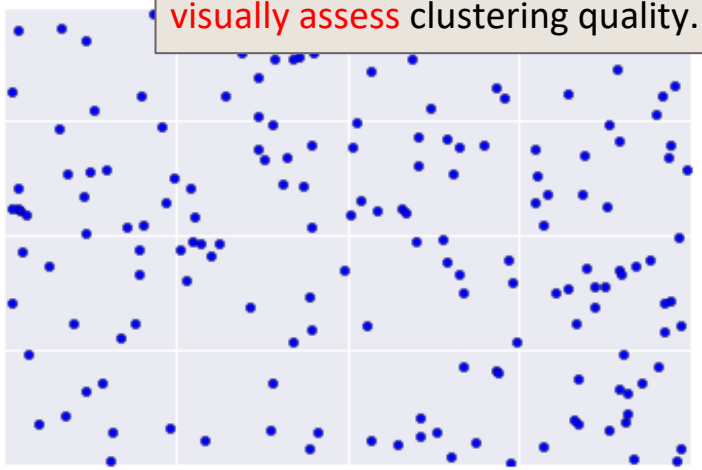
www.utm.my

Evaluating results

Clustering is unsupervised and no ground truth is available to verify results. The absence of truth can complicate assessments of quality.



The real-world datasets typically don't offer obvious clusters of examples.



The real-world data often looks more like this, making it difficult to visually assess clustering quality.

Evaluating results

There are several heuristics and best practices that you can iteratively apply to improve the quality of your clustering.

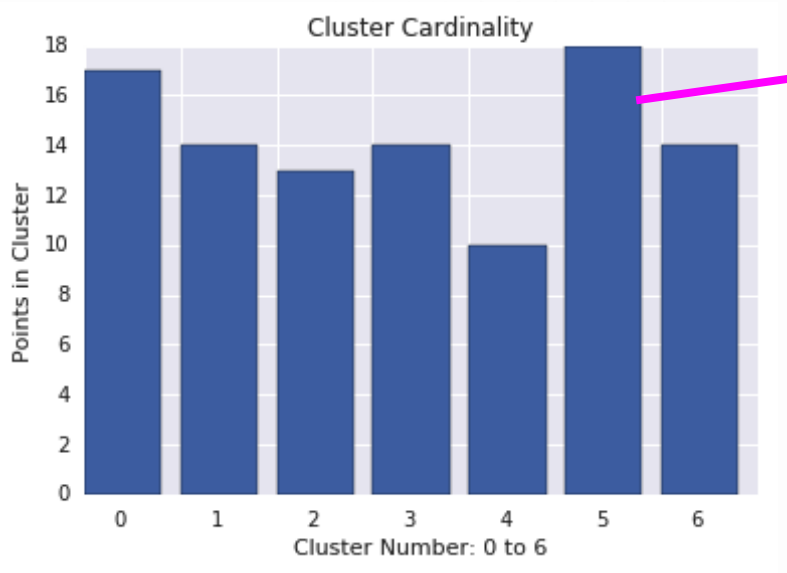
Step 1: Assess quality of clustering

Check the clusters look as you would expect, and the examples that you consider to be similar to each other appear in the same cluster. Commonly-used metrics are:

1. Cluster cardinality
2. Cluster magnitude
3. Downstream performance

Cluster cardinality

Cluster cardinality is the number of examples per cluster. Plot the cluster cardinality for all clusters and investigate clusters that are major outliers.

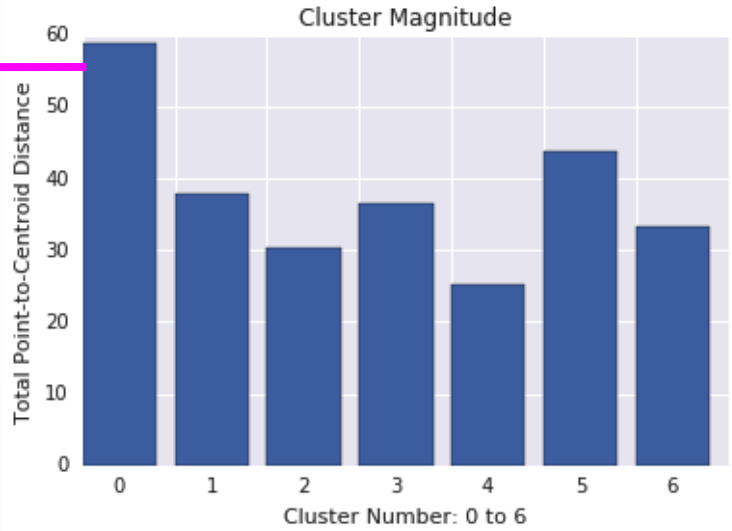


Cluster 5 are the major outliers.

Cluster Magnitude

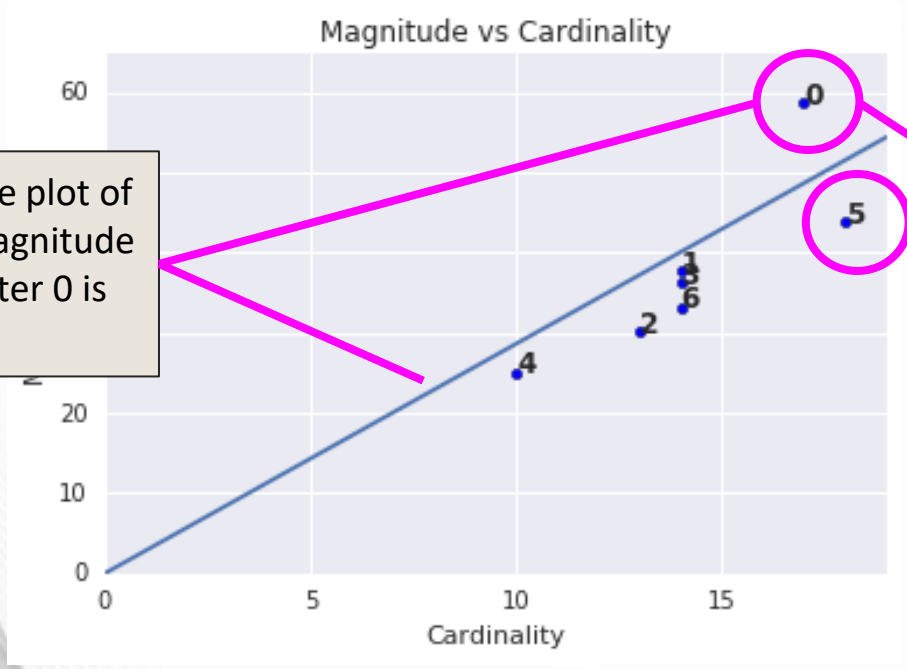
Cluster magnitude is the sum of distances from all examples in a cluster to the cluster's centroid. Plot cluster magnitude for all clusters and investigate outliers.

Cluster 0 is an outlier.



Magnitude versus Cardinality

Anomalous clusters can be identified by looking for ones where this relationship between cardinality and magnitude is very different than for other clusters.



Fitting a line to the plot of cardinality and magnitude suggests that cluster 0 is anomalous.

Cluster 5 is also far from the line, but if cluster 0 were omitted, the new fitted line would be much closer to cluster 5.

Evaluating results

Step 2: Find the optimal number of clusters

K-Means requires you to decide the number of clusters, k beforehand. To determine an optimal k, run the algorithm with increasing values of k and note the sum of all cluster magnitudes.

Above a certain k, the reduction in loss becomes marginal with increasing k.

Consider using the k where the slope first has a drastic change. This is known as the elbow method.



The optimal k is approximately 11.

As k increases, clusters become smaller, and the total distance of points from centroids decreases. This total distance can be treated as a loss.

Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Applications
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

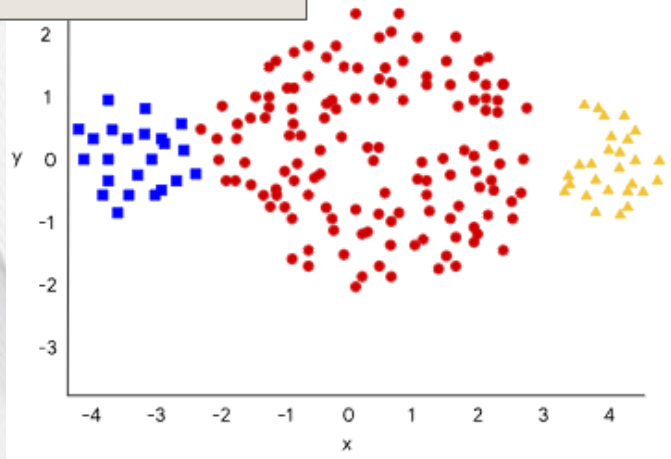
Advantage of K-means Clustering

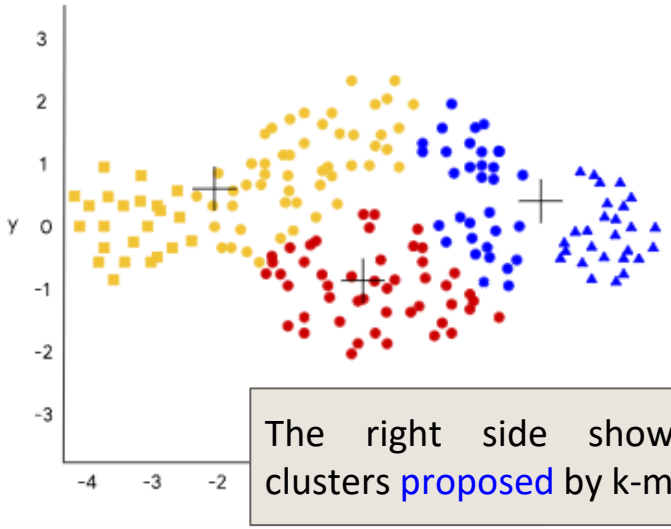
- It is relatively simple to implement.
- It scales to large data sets.
- It always converges.
- It allows warm-starting the positions of centroids.
- Smoothly adapts to new examples.
- Can be generalized to clusters of different shapes and sizes, such as elliptical clusters.

Generalizing K-means

A straightforward implementation of k-means can struggle with clusters of different densities and sizes.

The left side of the figure displays the clusters we'd expect to see.





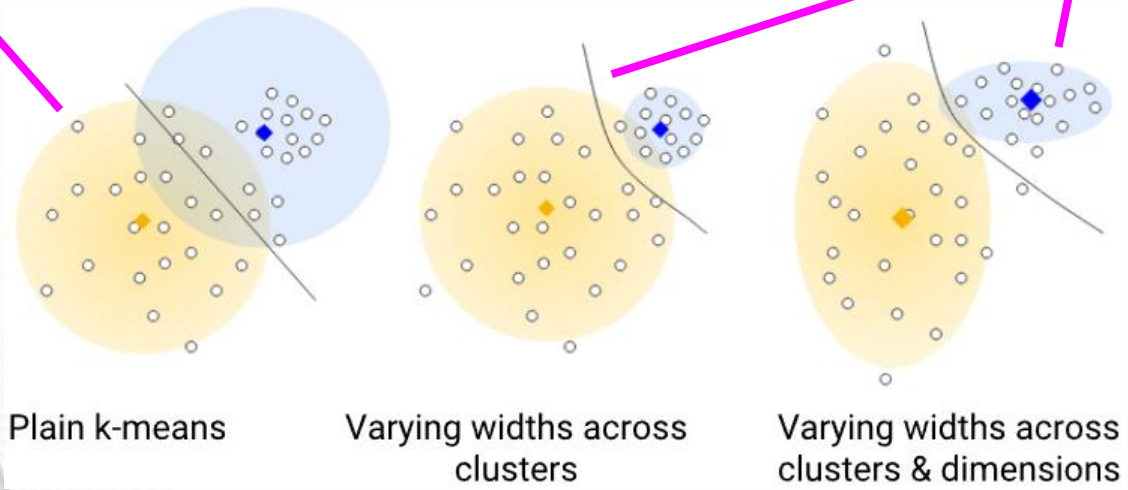
The right side shows the clusters proposed by k-means.

Generalizing K-means

For better performance on imbalance clusters like the ones shown in the previous figure, you can generalize by adapting k-means.

The first dataset shows K-Means without generalization.

The second and third dataset allow for clusters to vary in width.



Topic 13: Clustering

- Clustering: An introduction
- Clustering algorithms & Its Applications
- Types of clustering methods
- K-Means clustering
- Advantages of K-means clustering
- Disadvantages of K-Means clustering

www.utm.my

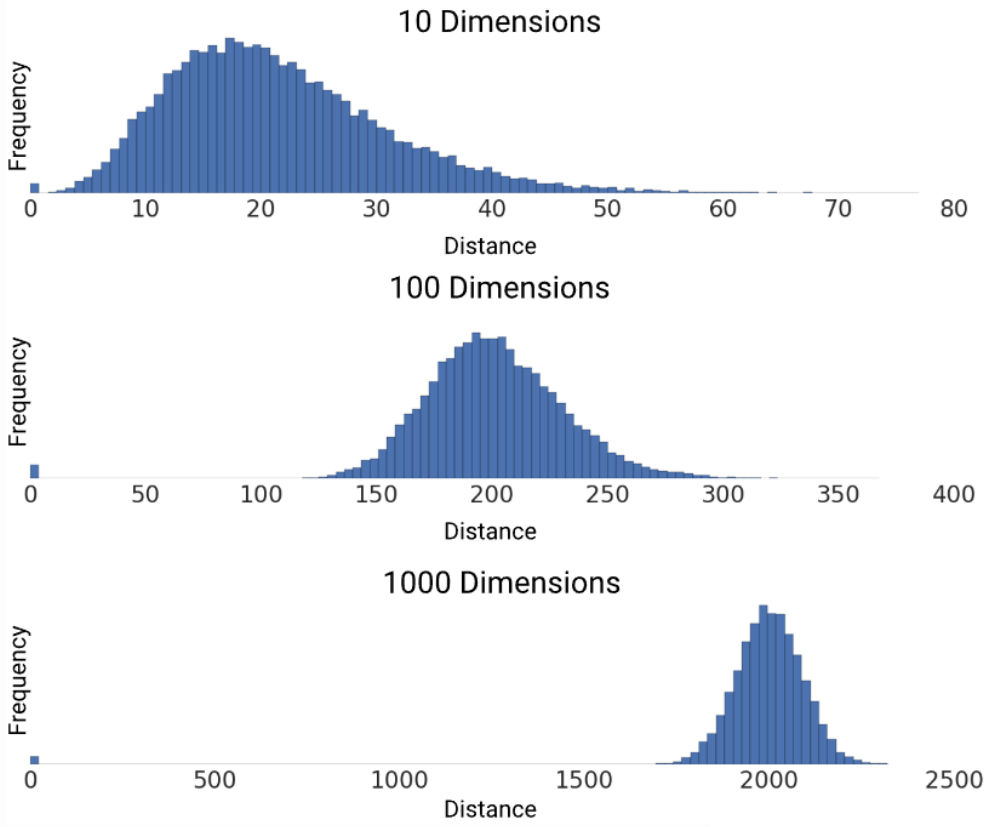
Disadvantage of K-means Clustering

- k must be chosen manually.
- Results depend on initial values. For low k, this dependence can be mitigated by running k-means several times with different initial values and picking the best result. As k increases, k-means seeding may needed to pick better initial centroids.
- Difficulty clustering data of varying size and densities without generalization.
- Difficulty clustering outliers. Centroids can be dragged by outliers, or outliers might get their own cluster instead of being ignored. Consider removing or clipping outliers before clustering.
- Difficulty scaling with number of dimensions. As the number of dimension in the data increases, a distance-based similarity measure converges to a constant value between any given examples. PCA or spectral clustering can be used to reduce dimensionality.

Curse of dimensionality and spectral clustering

As the dimensions increase, the standard deviation in distance between examples shrinks relative to the mean distance between examples.

This convergence means that k-means becomes less effective at distinguishing between examples as the dimensionality of the data increases. (curse of dimensionality)



Curse of dimensionality and spectral clustering

This diminishing in performance can be prevented by using spectral clustering, which adds pre-clustering steps to the algorithm. To perform spectral clustering:

1. Reduce the dimensionality of feature data by using PCA.
2. Project all data points into the lower-dimensional subspace.
3. Cluster the data in this subspace using your chosen algorithm.